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CS-499

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3-2 Milestone Two: Enhancement One: Software Design and Engineering

The original IT-145 project included a placeholder `reserveAnimal()` method that did not contain any functional logic. It simply printed a message saying the animal was reserved, without checking the animal type, verifying availability, or updating the reservation status. Because of this, the method did not meet the requirements for real functionality or demonstrate software engineering principles such as conditional logic, data validation, or object state updates.

For my Software Design & Engineering enhancement, I fully implemented the `reserveAnimal()` method. The enhanced version now:

- Accepts user input for animal type and in-service country
- Searches the correct list (`dogList` or `monkeyList`)
- Validates that the animal is:
 - fully trained (“in service”)
 - not already reserved
 - located in the requested country
- Updates the object’s reserved status
- Provides clear user feedback for success, invalid input, or no available animals

This enhancement replaces the placeholder method with a complete, functional implementation that demonstrates real decision-making and state management.

This enhancement improves the artifact by adding meaningful logic and functionality that was missing from the original project. Instead of a static print statement, the method now:

- Uses conditional branching
- Iterates through data structures
- Performs validation checks

- Updates object attributes
- Handles multiple error cases
- Supports both dogs and monkeys

These improvements make the program behave like a real reservation system and demonstrate stronger software engineering skills.

This enhancement supports the following CS-499 course outcomes:

- Software Design & Engineering: I improved an existing program by adding functional logic, validating user input, and updating object state. The enhancement shows understanding of control flow, data structures, and object-oriented programming.
- Algorithms & Data Structure Use: The method uses iteration and conditional checks to search lists and identify valid animals.

This enhancement helped me strengthen my understanding of how to modify existing codebases, integrate new logic into an established structure, and ensure that new features interact correctly with existing classes and data. It also reinforced the importance of testing with different scenarios to confirm that the logic works for all cases.